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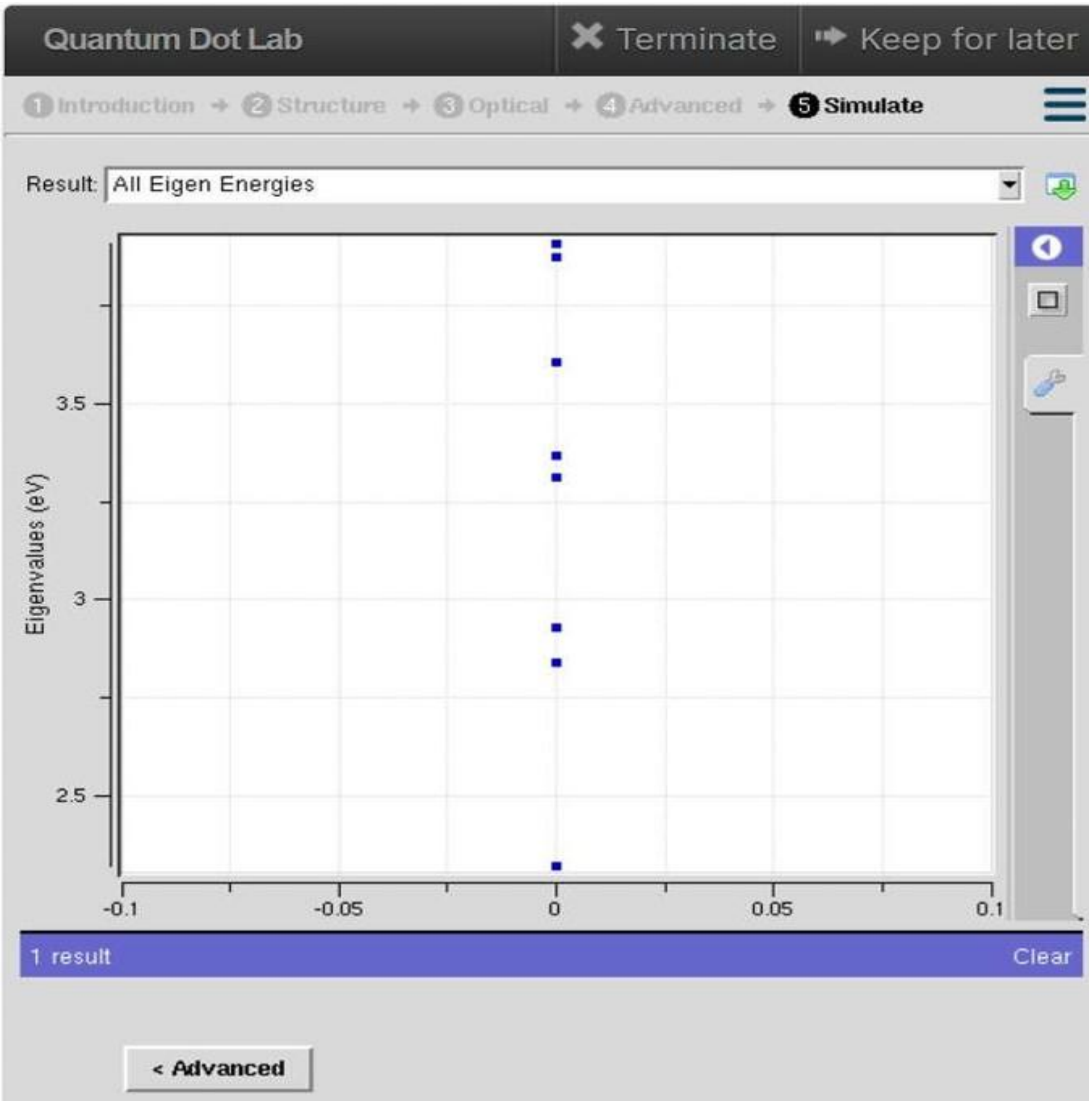
SLOT C- ECA3102 - NANO ELECTRONICS

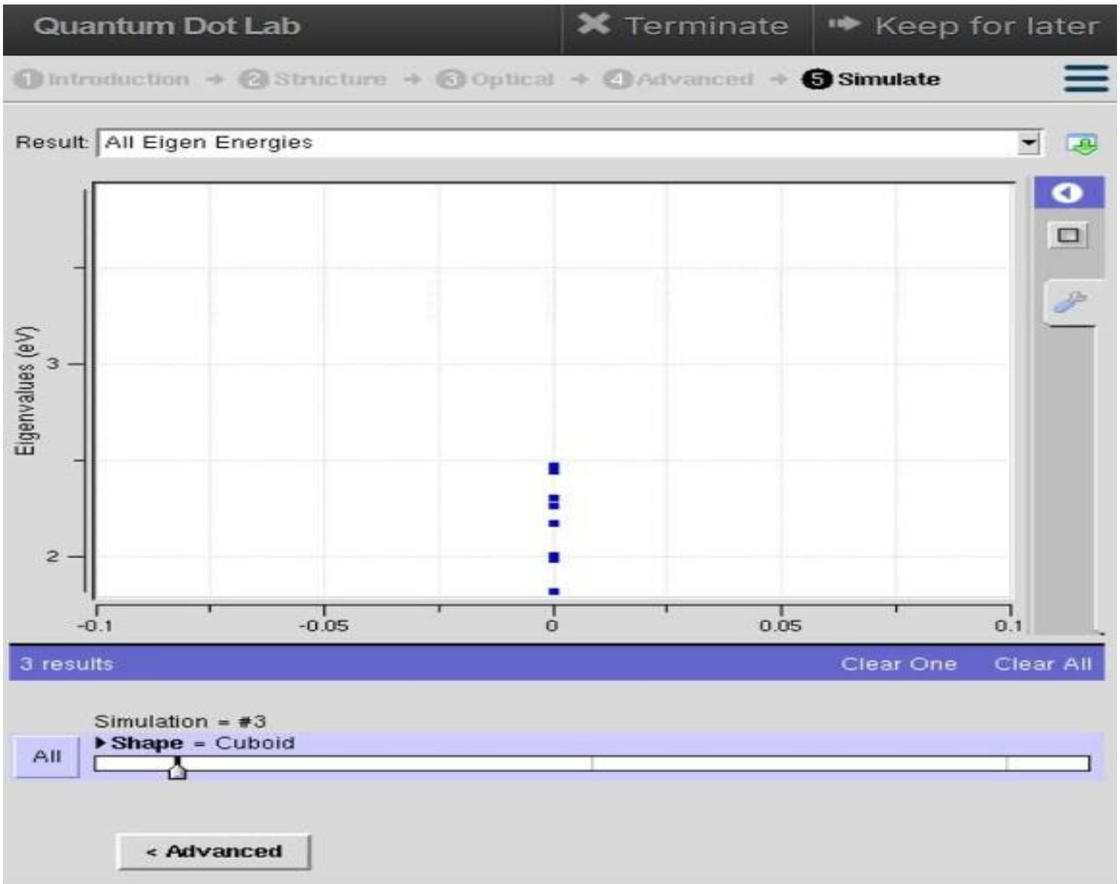
EXPERIMENT 3- SIMULATION OF ENERGY STATES OF QUAANTUM DOT USING NANOHUB

Aim:
To obtain the energy states of Quantum dots



EXPLANATION:

The simulation shows that the quantum dot has **discrete eigen energy levels** due to quantum confinement. These quantized energy states determine the optical and electronic properties of the quantum dot, making it useful for **LEDs, lasers, solar cells, and quantum computing applications**.



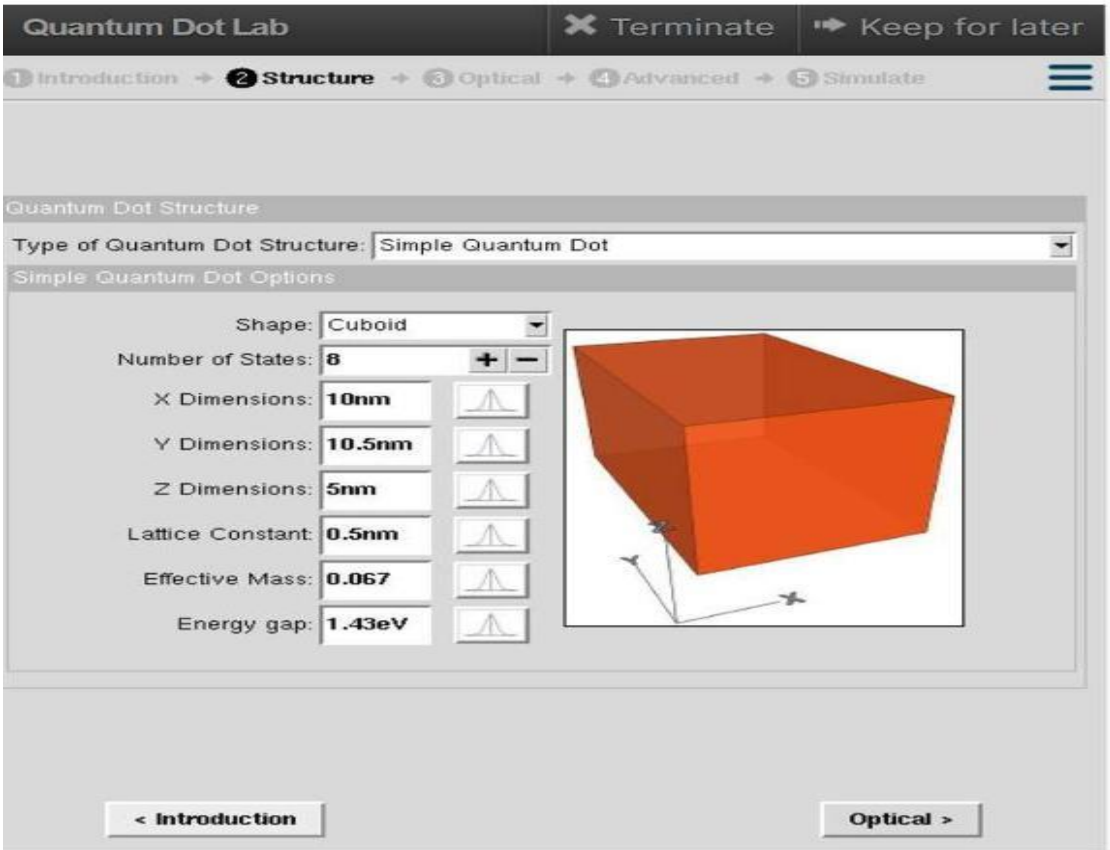
Effect of quantum dot size

- Smaller dot → larger energy spacing → higher energy (blue shift).
- Larger dot → smaller energy spacing → lower energy (red shift).

About the “Shape = Cuboid”

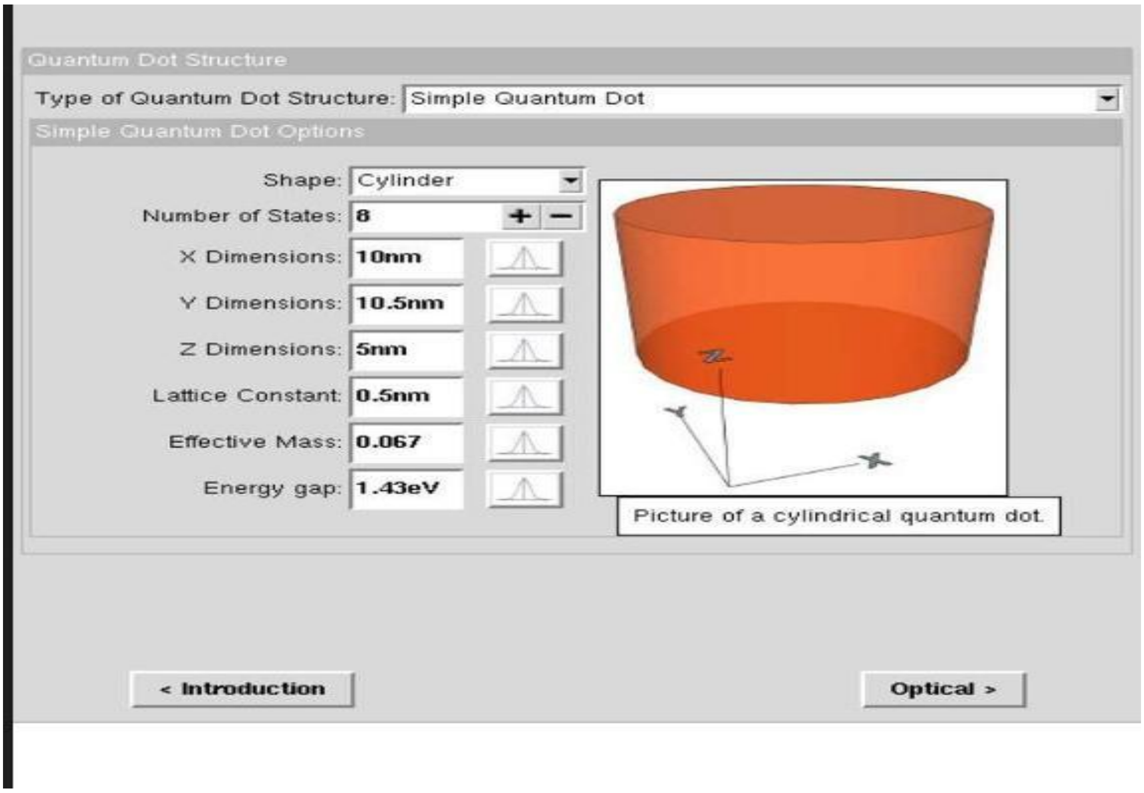
The selected quantum dot shape is Cuboid. Confinement occurs along length, width, and height, so the energy levels depend on all three dimensions.

The simulation confirms quantum confinement in a cuboid quantum dot by showing discrete eigen energy levels in the range of approximately 1.8–2.5 eV.



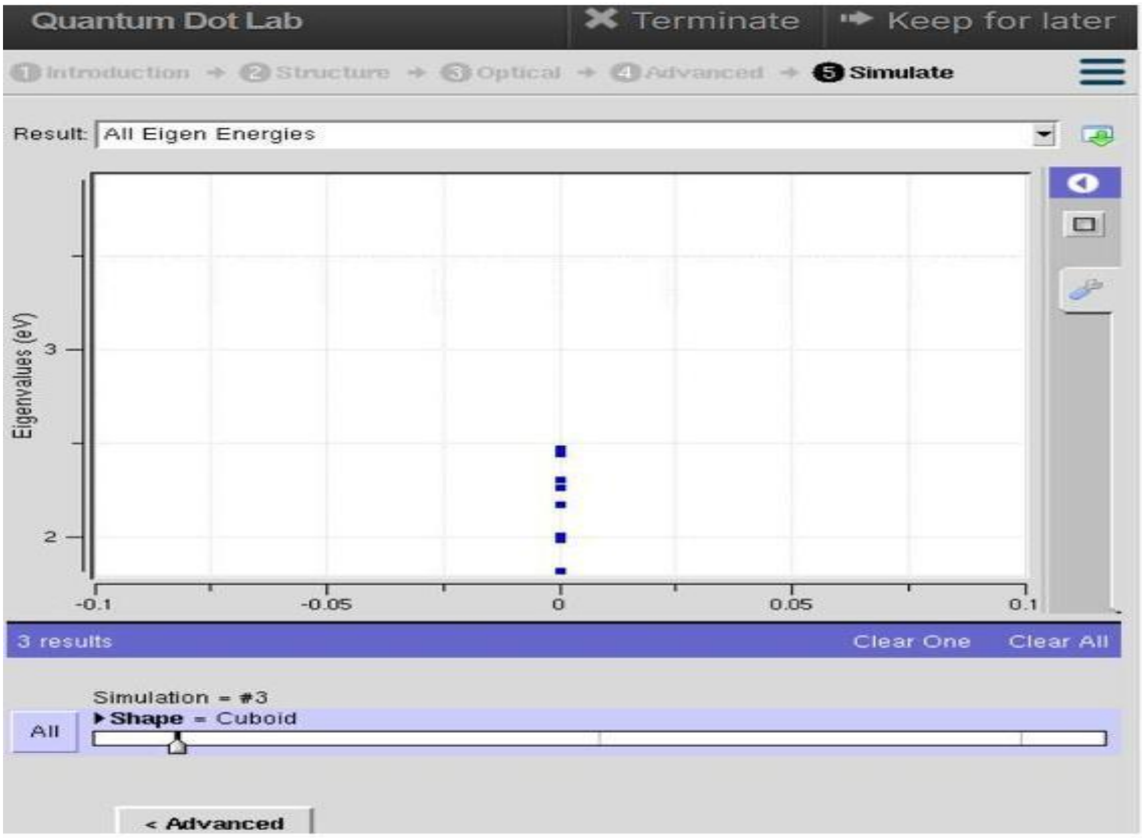
EXPLANATION:

The structure defines a cuboid quantum dot with specified dimensions and material properties. These parameters control electron confinement, resulting in quantized energy levels and influencing the electronic and optical behavior of the quantum dot.



EXPLANATION:

The simulation uses a **cylindrical quantum dot** with specified dimensions and material properties. The cylindrical confinement produces **quantized energy levels**, making the structure suitable for studying the electronic and optical behavior of quantum devices.



Observations from the graph

- The lowest energy level is around 1.8 eV → Ground state.
- Higher levels around 2.0–2.5 eV → Excited states.
- The unequal spacing between points indicates different quantum states of the confined electron.

Physical interpretation

- Lower energy → more stable state.
- Higher energy → excited electron state.
- Energy gap between levels determines optical properties such as absorption and emission wavelength.

Effect of size

- Smaller cuboid → larger energy spacing → higher energy (blue shift).
- Larger cuboid → smaller energy spacing → lower energy (red shift).

CONCLUSION:

The simulation of the cylindrical quantum dot demonstrates that electrons are confined in all three dimensions, resulting in discrete (quantized) energy levels. The size and shape of the quantum dot significantly influence its electronic and optical properties, making it suitable for nanoelectronic and optoelectronic applications